FSCT 8561 — Lab 2 Rubric: Network Scanning & Enumeration

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Total points: 100

Component	Criteria	Excellent — Full Credit	Partial Credit	No / Minimal Credit	Points
Environment Setup & Readiness	Python 3, Nmap, and python-nmap installed and functional	Environment correctly configured; Python sockets work; Nmap is accessible; nmap.PortScanner() initializes successfully	Minor setup or configuration issues	Environment is not functional or required tools are missing	**5**
Raw Python Port Scanner (scanner.py)	TCP Connect scanning using socket, connect_ex(), timeout, loop, and proper socket closing	Correctly scans the selected port range; uses a new socket appropriately; identifies open ports; sockets are closed; implementation is clear and well structured	Scanner generally works but has minor logic, structure, timeout, or resource-handling issues	Scanner is incomplete, largely incorrect, or non-functional	**20**
Input Validation & Robustness	Target resolution, port-range validation, error handling, controlled scan range	Handles invalid hostname/IP, invalid/non-numeric ports, invalid ranges, no open ports, and connection errors/timeouts gracefully with informative messages	Most cases handled but some errors can cause incorrect output or termination	Little or no validation/error handling	**15**
Service Identification & Results	Open-port collection, likely service identification, clear reporting	Correctly reports discovered open ports and likely service names when available; results are clear and readable; appropriately avoids claiming that every unsuccessful connection is “filtered”	Results mostly correct but incomplete, unclear, or occasionally over-interpreted	Results missing or substantially incorrect	**10**

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python-nmap Scanner (nmap_scanner.py)	Uses PortScanner, performs authorized scan, programmatically extracts results	Correctly uses python-nmap; accepts/validates target and range; extracts and clearly displays port state and available service information rather than simply printing raw results	Functional implementation but limited result processing, validation, or formatting	Missing, non-functional, or only prints raw Nmap data without meaningful processing	**15**
Scanner Comparison & Technical Understanding	Comparison of raw sockets and python-nmap	Clearly explains differences in code complexity, abstraction, available information, capabilities/limitations, and why building the raw version first is educationally useful	Comparison demonstrates general understanding but lacks technical depth or misses some questions	Comparison missing or demonstrates substantial misunderstanding	**10**
Testing & Demonstration	Authorized target, controlled range, robustness tests, both implementations demonstrated	Recording clearly demonstrates raw scanner, python-nmap scanner, known open port where available, selected range, required robustness tests, and comparison	Demonstration is mostly complete but some required tests/evidence are missing	Recording missing or provides insufficient evidence that implementations work	**10**
Reflection & Security Analysis	Reflection questions + 300–400 word analysis based on actual results	Accurate and thoughtful responses; analysis explicitly connects actual ports/services to attack surface, risks, scanner limitations, authorization, and practical defensive measures	Generally correct but mostly descriptive, weakly connected to actual results, or lacks depth	Missing, inaccurate, or demonstrates limited understanding	**15**